

FINITE-DEFECT PATH MODELS FOR PERIPHERAL BENZEL TILINGS: PROPP'S PROBLEM 6 AND THE FIRST DEFECT DIAGONAL

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ABSTRACT. Propp asked for the number of tilings of the peripheral benzel $\mathcal{B}(n, 2n - 3)$ by right stones and all three orientations of bones. We first prove his conjectured formula

$$T_{103}(n, 2n - 3) = \frac{(3n + 3)(3n - 7)!}{(n - 5)!(2n - 1)!} \quad (n \geq 5).$$

The proof partitions the cells into periodic three-cell owners. A linear owner-label energy turns every tiling into a directed Y of three labelled ballot paths, whose two sink chiralities are evaluated by a three-variable constant term and Lagrange inversion. We then place this rigid case in a finite-defect hierarchy on the diagonals $b = 2a - d$. In the nonoverlap range $2k \leq a - 1$, the exact defect number is $\binom{k}{2}$ for $d = 3k$ and $\binom{k+1}{2}$ for $d = 3k + 1$. Consequently $d = 4$ is the first one-defect diagonal. Deleting its unique defect gives three independent labelled ballot paths and five defect classes. A literal inverse construction and a specialized Lagrange–Good evaluation yield an algebraic ordinary generating function for every $T_{103}(a, 2a - 4)$.

1. INTRODUCTION

Benzels are threefold-symmetric regions in the hexagonal grid introduced by Propp as finite regions for exact trimer enumeration [7]. The five relevant trihexes are two triangular *stones* and three collinear *bones*. Write $T_{ijk}(a, b)$ for the number of tilings of the (a, b) -benzel when right stones are allowed if $i = 1$, left stones are allowed if $j = 1$, and k bone orientations are allowed.

Several of Propp's problems have been solved using abacus and compression bijections [3, 2]. Those methods are particularly effective when at least one bone orientation is forbidden. The present problem permits all three bone orientations and lies on the peripheral diagonal $b = 2a - 3$. Propp states the following as Problem 6 and reports verification for $5 \leq n \leq 16$ [7]. Propp's online status table, whose most recent problem update is dated June 2025, still listed Problem 6 as open when accessed on 22 August 2026 [8].

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Theorem 1.1 (Propp’s Problem 6). *For every integer $n \geq 5$,*

$$T_{103}(n, 2n - 3) = \frac{(3n + 3)(3n - 7)!}{(n - 5)!(2n - 1)!}.$$

The proof uses a periodic right-stone tiling to partition the plane into three-cell owners. Although the benzel has quadratic area, the Conway–Lagarias invariant and an owner-label energy force every admissible tiling to differ from this periodic background along only three directed paths. These paths have total length $3n - 6$ and meet at a single interior owner. The rest of the proof enumerates the three-path configurations.

Put $m = n - 5$. The conjectured number has the useful ballot form

$$(1) \quad 2 \binom{3m + 8}{m} - \binom{3m + 8}{m - 1} = \frac{3(m + 6)}{2m + 9} \binom{3m + 8}{m}.$$

The path reduction explains both terms: one sink chirality gives a cyclically telescoping product of binomial counts, while the other gives ballot counts.

2. BENZELS, STONES, AND BONES

Represent a hexagonal cell center by $(i, j) \in \mathbb{Z}^2$ and put $k = 1 - i - j$. This is the barycentric convention used by Propp [7]. The (a, b) -benzel is the cell set

$$\mathcal{B}(a, b) = \{(i, j) : -(a - 1) \leq j - i, k - j, i - k \leq b - 1\}.$$

The right stone is the translate class of

$$R = \{(0, 0), (1, 0), (0, 1)\},$$

and the three bones have direction sets

$$H = \{(0, 0), (1, 0), (2, 0)\},$$

$$V = \{(0, 0), (0, 1), (0, 2)\},$$

$$D = \{(0, 0), (1, -1), (2, -2)\}.$$

Only translates, not rotations beyond these three listed bone types, are used.

The area and Conway–Lagarias invariant give the following tile counts [1, 7].

Lemma 2.1. *Every type-103 tiling of $\mathcal{B}(n, 2n - 3)$ has*

$$\frac{(n - 5)(n - 2)}{2} \text{ right stones and } 3(n - 2) \text{ bones.}$$

Proof. Substitution in the benzel area formula gives

$$|\mathcal{B}(n, 2n - 3)| = \frac{3(n - 2)(n + 1)}{2},$$

so a tiling has $(n - 2)(n + 1)/2$ trihexes. Since no left stones are allowed, the Conway–Lagarias invariant equals the number of right stones. Its $a + b \equiv 0$

(mod 3) formula at $(a, b) = (n, 2n - 3)$ is

$$\frac{3(a - b)^2 - a - b}{6} = \frac{(n - 5)(n - 2)}{2}.$$

Subtracting from the total number of trihexes gives $3(n - 2)$ bones. $\square$

3. PERIODIC OWNERS AND THE ENERGY CERTIFICATE

Put $t = n - 2$ and work modulo three. An *owner anchor* is a pair (q, r) with

$$q - r \equiv t \pmod{3}.$$

The anchor owns the three cells

$$(q, r), \quad (q + 1, r), \quad (q, r + 1),$$

labelled 0, 1, 2, respectively. Every cell of the infinite hexagonal grid has a unique owner: among its three possible right-stone anchors, the three values of $q - r$ represent the three residue classes.

For an owner anchor (q, r) define

$$(2) \quad u = \frac{t - q + r}{3}, \quad v = \frac{t - q - 2r}{3}, \quad w = \frac{t + 2q + r}{3}.$$

These are integers with $u + v + w = t$.

Lemma 3.1 (Owner simplex). *The owners meeting $\mathcal{B}(n, 2n - 3)$ are in bijection with*

$$\Delta_t = \{(u, v, w) \in \mathbb{N}_0^3 : u + v + w = t\}.$$

Every non-corner owner contributes all three of its cells to the benzel. The three corner owners contribute two cells each: $(0, t, 0)$ misses label 0, $(0, 0, t)$ misses label 1, and $(t, 0, 0)$ misses label 2.

Proof. For a cell (i, j) put $k = 1 - i - j$ as above. Substitution of the three labels at (q, r) gives the following ordered triples $(j - i, k - j, i - k)$:

label	$j - i$	$k - j$	$i - k$
0	$3u - t$	$3v - t + 1$	$3w - t - 1$
1	$3u - t - 1$	$3v - t$	$3w - t + 1$
2	$3u - t + 1$	$3v - t - 1$	$3w - t$

The benzel bounds are $-t - 1 \leq j - i, k - j, i - k \leq 2t$. If any one of the three cells lies in the benzel, its row in the table, together with the integrality of u, v, w , forces $u, v, w \geq 0$. Conversely, suppose $u, v, w \geq 0$ and $u + v + w = t$. The table then shows

$$\begin{aligned} \text{label 0 is present} &\iff v < t, \\ \text{label 1 is present} &\iff w < t, \\ \text{label 2 is present} &\iff u < t. \end{aligned}$$

The inverse coordinate change is

$$q = w - u, \quad r = u - v;$$

indeed $q - r = t - 3u \equiv t \pmod{3}$. Thus every owner meeting the benzel corresponds to a unique point of Δ_t , and every point of Δ_t contributes at least two cells. A coordinate can equal t only when the other two coordinates vanish. The three such points give exactly the three missing labels stated in the lemma; all other owners contribute all three cells. $\square$

At anchor (q, r) give labels 0, 1, 2 the respective integer energies

$$(3) \quad h_0(q, r) = q + r, \quad h_1(q, r) = -q, \quad h_2(q, r) = -r.$$

Lemma 3.2 (Local energy table). *Relative to the owner phase above, the four possible tile classes have the following energies.*

tile class	number of owners	energy
<i>in-phase right stone</i>	1	0
<i>wrong-phase right stone</i>	3	3
<i>two-owner bone</i>	2	1
<i>three-owner bone</i>	3	4

For a two-owner bone, the owner containing two cells is its source. The label missing at the source equals the label of the single target cell.

Proof. For an in-phase stone, the three energies at one owner sum to zero. The remaining cases depend only on the translation class modulo three. For a translate based at (i, j) , put

$$\rho \equiv i - j - t \pmod{3}, \quad \rho \in \{0, 1, 2\}.$$

Direct substitution in (3) gives

ρ	R	H	V	D
0	0	1	1	1
1	3	4	1	4
2	3	1	4	1.

Here the entry for R at $\rho = 0$ is the in-phase stone. The six energy-one bone cases have the following source-missing label and directed owner displacement:

bone and phase	label	step
$(H, 0)$	2	A
$(H, 2)$	2	B
$(V, 0)$	1	C
$(V, 1)$	1	B
$(D, 0)$	0	C
$(D, 2)$	0	A .

In every row, the two source cells are precisely the two labels other than the displayed one, and the target cell has the displayed label. Hence its energy is $-h_\ell(\text{source}) + h_\ell(\text{target}) = 1$. The three unlisted bone phases meet three owners and have energy four, which completes the table. $\square$

Proposition 3.3 (Energy rigidity). *Every right stone in a type-103 tiling of $\mathcal{B}(n, 2n - 3)$ is in phase, and every bone is a two-owner, label-preserving directed edge.*

Proof. Every full owner has total energy zero. At each corner the two present labels have total energy t , so Lemma 3.1 gives total benzel energy $3t = 3(n - 2)$.

Let s be the number of wrong-phase stones and f the number of three-owner bones. Lemma 2.1 gives $3(n - 2)$ bones. Of these, $3(n - 2) - f$ have energy one and f have energy four, while the s wrong-phase stones have energy three. The total tile energy is therefore

$$[3(n - 2) - f] + 4f + 3s = 3(n - 2) + 3(s + f).$$

Comparison with the benzel energy gives $s + f = 0$, hence $s = f = 0$. $\square$

4. THE DIRECTED Y

In anchor coordinates, the only directed displacements are

$$A = (2, -1), \quad B = (-1, -1), \quad C = (-1, 2).$$

In simplex coordinates they are

$$\begin{aligned} A &: (u, v, w) \mapsto (u - 1, v, w + 1), \\ B &: (u, v, w) \mapsto (u, v + 1, w - 1), \\ C &: (u, v, w) \mapsto (u + 1, v - 1, w). \end{aligned}$$

Label 0 permits A, C , label 1 permits B, C , and label 2 permits A, B . The corresponding potential h_ℓ increases by one on every edge carrying label ℓ . We replace each bone by the directed edge from its two-cell owner to its one-cell owner; the second table in the proof of Lemma 3.2 shows that this correspondence is unique in both directions.

Proposition 4.1 (Directed- Y bijection). *Type-103 tilings of $\mathcal{B}(n, 2n - 3)$ are in bijection with triples of labelled paths in Δ_t satisfying the following conditions.*

- (1) *The paths of labels 0, 1, 2 start at $(0, t, 0)$, $(0, 0, t)$, and $(t, 0, 0)$, respectively.*
- (2) *Each path uses its two permitted steps, and the vertex sets of two different paths intersect only at their common endpoint.*
- (3) *The paths have one common endpoint (u, v, w) with $u, v, w > 0$.*

Every full owner outside the three paths is occupied by a right stone.

Proof. An outgoing edge uses two cells of its source owner, whereas an incoming edge uses one cell of its target owner. At a full owner not occupied by a stone, let o and i denote the numbers of outgoing and incoming edges. Exact coverage gives $2o + i = 3$, and two outgoing edges would overlap. Hence either $(o, i) = (1, 1)$ or $(o, i) = (0, 3)$. In the first case the incoming edge covers the label omitted by the outgoing edge, so both carry the same label.

In the second case the incoming labels are 0, 1, 2. At a deficient corner the same argument gives $2o + i = 2$: either there is one outgoing edge carrying the missing label, or there are two incoming edges carrying the two present labels.

Since $|\Delta_t| = \binom{t+2}{2} = \binom{n}{2}$, Lemma 2.1 leaves

$$\binom{n}{2} - \frac{(n-5)(n-2)}{2} = 3t + 1$$

owners not occupied by stones. There are $3t$ bone edges. If K is the number of full three-incoming sinks and P the number of two-incoming corner sinks, every other active owner has one outgoing edge. Counting outgoing edges therefore gives

$$3t = (3t + 1) - K - P,$$

so $K + P = 1$.

No directed cycle is possible because the constant label potential strictly increases along every edge of a pass-through component. It remains to rule out $P = 1$. By cyclic symmetry suppose $(0, t, 0)$ is the corner sink. A label-1 path from $(0, 0, t)$ to this corner uses only B -steps: B leaves u unchanged, C increases it, and both endpoints have $u = 0$. Its vertex set is therefore

$$\{(0, b, t - b) : 0 \leq b \leq t\}.$$

A label-2 path from $(t, 0, 0)$ to the same corner must use exactly t A -steps and t B -steps. Immediately after its final A -step it is at $(0, b, t - b)$ for some $b < t$; equality $b = t$ would make the preceding w -coordinate negative. This vertex belongs to the label-1 path and is not the common corner sink, contradicting the local coverage rules. Hence $K = 1$ and $P = 0$.

Following outgoing edges from the three corners now gives the required three paths. They cannot meet before the unique sink: a pass-through owner has only one incoming edge, and the sink already has its three incoming labels. Their common endpoint is a full owner and therefore has three positive simplex coordinates. Any further component would have no sink and hence, by finiteness, would contain a directed cycle.

Conversely, replace every permitted labelled step by the unique bone from Lemma 3.2. At a source, the outgoing bone covers the two present labels. At a pass-through owner, the incoming bone covers the path label and the outgoing bone covers the other two labels. At the common sink, the three incoming bones cover labels 0, 1, 2. In-phase stones cover every unused full owner. The intersection condition ensures that these local coverings are compatible, so they form a unique tiling. $\square$

Write the sink as

$$(u, v, w) = (x + 1, y + 1, z + 1), \quad x + y + z = m = n - 5.$$

At the sink the last directions must be distinct, since equal directions would give the same predecessor owner to two paths. Respecting the two allowed

directions for each label leaves exactly two possibilities:

$$(+): (C, B, A), \quad (-): (A, C, B)$$

for labels $(0, 1, 2)$.

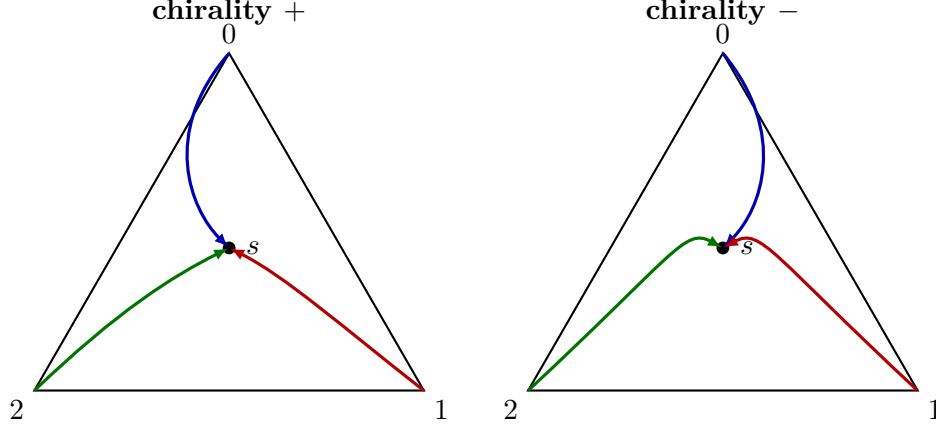


FIGURE 1. The three labelled arms, the fixed sink, and the two possible last-edge chiralities. The drawing is schematic; the proof uses the simplex coordinates.

5. FIXED-SINK ENUMERATION

Put

$$N_x = 2x + y + 2, \quad N_y = 2y + z + 2, \quad N_z = 2z + x + 2,$$

and adopt the convention $\binom{N}{-1} = 0$.

Lemma 5.1 (Independent arms). *Fix the sink $(x+1, y+1, z+1)$ and one of the two last-edge chiralities. Any three individually admissible labelled arms with those endpoints and last edges intersect only at the sink. The numbers of choices for the arms of labels $(0, 1, 2)$ are, respectively,*

$$\begin{aligned} (+): & \quad \binom{N_z}{z+1} - \binom{N_z}{z}, \quad \binom{N_x}{x+1} - \binom{N_x}{x}, \quad \binom{N_y}{y+1} - \binom{N_y}{y}, \\ (-): & \quad \binom{N_z}{z} - \binom{N_z}{z-1}, \quad \binom{N_x}{x} - \binom{N_x}{x-1}, \quad \binom{N_y}{y} - \binom{N_y}{y-1}. \end{aligned}$$

Proof. The step multiplicities from the three sources to the sink are

label	step multiplicities	
0	$A : z + 1$	$C : x + z + 2$
1	$B : x + y + 2$	$C : x + 1$
2	$A : y + z + 2$	$B : y + 1$.

For label 0, the only nontrivial simplex constraint along the path is $u \geq 0$, or equivalently that every prefix contains at least as many C -steps as A -steps.

For labels 1 and 2, the corresponding conditions are, respectively, at least as many B -steps as C -steps and at least as many A -steps as B -steps. Remove the prescribed last step. The reflection principle for a word with k minority letters gives $\binom{N}{k} - \binom{N}{k-1}$: reflecting the initial segment ending at the first negative prefix bijects the bad words with the words having $k - 1$ minority letters. This yields the six displayed counts.

It remains to check that independently chosen arms cannot meet prematurely. Every vertex on the label-0, label-1, and label-2 arms satisfies, respectively,

$$v \geq y + 1, \quad w \leq z + 1; \quad w \geq z + 1, \quad u \leq x + 1; \quad u \geq x + 1, \quad v \leq y + 1.$$

Thus a common vertex of arms 0 and 1 must lie on $w = z + 1$. In positive chirality the final B -step of arm 1 is its first arrival at this boundary; in negative chirality the final A -step of arm 0 is its first arrival. Hence the two arms meet only at the sink. The same argument on the boundaries $u = x + 1$ and $v = y + 1$, cyclically, separates the other two pairs. This proves both independence and the stated arm counts. $\square$

Corollary 5.2. *Let $P(x, y, z)$ and $Q(x, y, z)$ denote the positive- and negative-chirality counts at the fixed sink. Then*

$$(4) \quad P(x, y, z) = \prod_{\text{cyc}} \binom{2x + y + 2}{x},$$

$$(5) \quad Q(x, y, z) = \prod_{\text{cyc}} \left(\binom{2x + y + 2}{x} - \binom{2x + y + 2}{x - 1} \right).$$

In particular,

$$(6) \quad \frac{Q(x, y, z)}{P(x, y, z)} = \frac{(x + 3)(y + 3)(z + 3)}{(x + y + 3)(y + z + 3)(z + x + 3)}.$$

Proof. The negative-chirality product in Lemma 5.1 is exactly (5). For $N = 2r + s + 2$,

$$\binom{N}{r + 1} - \binom{N}{r} = \frac{s + 1}{r + 1} \binom{N}{r}.$$

Applying this identity to the three positive-chirality factors gives the additional multiplier

$$\frac{x + 1}{z + 1} \frac{y + 1}{x + 1} \frac{z + 1}{y + 1} = 1,$$

which proves (4). Finally,

$$\frac{\binom{2r+s+2}{r} - \binom{2r+s+2}{r-1}}{\binom{2r+s+2}{r}} = \frac{s + 3}{r + s + 3}.$$

Multiplying its three cyclic instances gives (6). $\square$

6. SUMMING THE SINKS

Define

$$P(t) = \sum_{x,y,z \geq 0} P(x,y,z)t^{x+y+z}, \quad Q(t) = \sum_{x,y,z \geq 0} Q(x,y,z)t^{x+y+z}.$$

Proposition 6.1. *Let $T = T(t)$ be the unique formal power series satisfying*

$$T = 1 + tT^3.$$

Then

$$(7) \quad P(t) = \frac{T^9}{(3-2T)(T^2-3T+3)},$$

$$(8) \quad Q(t) = (2-T)^3 P(t),$$

$$(9) \quad P(t) + Q(t) = T^9 \frac{3-T}{3-2T}.$$

Proof. Use

$$\binom{2x+y+2}{x} = [a^x](1+a)^{2x+y+2}$$

and its cyclic analogues in (4). Summing three geometric series gives (10): the powers of x , for example, form the geometric ratio $t(1+a)^2(1+c)/a$. Thus

$$(10) \quad P(t) = \text{CT}_{a,b,c} \frac{(1+a)^2(1+b)^2(1+c)^2}{D_a D_b D_c},$$

where

$$\begin{aligned} D_a &= 1 - \frac{t(1+a)^2(1+c)}{a}, \\ D_b &= 1 - \frac{t(1+b)^2(1+a)}{b}, \\ D_c &= 1 - \frac{t(1+c)^2(1+b)}{c}. \end{aligned}$$

We evaluate this formal constant term by the multivariate Lagrange–Good residue formula [4]. Its small-root equations are

$$a = t(1+a)^2(1+c), \quad b = t(1+b)^2(1+a), \quad c = t(1+c)^2(1+b).$$

They have a unique solution in $t\mathbb{Q}[[t]]^3$: once all coefficients below degree d are known, the equations determine the degree- d coefficients. Cyclic symmetry therefore gives $a = b = c = U$, where $U = t(1+U)^3$; put $T = 1 + U$.

At the small root, put

$$r = \frac{U}{T} = \frac{T-1}{T}, \quad d = 1 - 2r = \frac{2-T}{T}.$$

For $\phi_a = (1+a)^2(1+c)$ and its cyclic analogues, the Good denominator is the determinant of

$$\left(\delta_{ij} - \frac{x_j}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right)_{i,j \in \{a,b,c\}}.$$

At the small root this matrix and its determinant are

$$J = \begin{pmatrix} d & 0 & -r \\ -r & d & 0 \\ 0 & -r & d \end{pmatrix}, \quad \det J = d^3 - r^3 = \frac{(3-2T)(T^2-3T+3)}{T^3}.$$

The numerator in (10) becomes T^6 . Division by the determinant contributes the additional factor T^3 , proving (7).

The ballot identity

$$\binom{N}{k} - \binom{N}{k-1} = [a^k](1-a)(1+a)^N$$

with $\binom{N}{-1} = 0$ shows that the constant term for $Q(t)$ has the additional numerator $(1-a)(1-b)(1-c)$. At the small root this is $(1-U)^3 = (2-T)^3$, proving (8). Finally,

$$1 + (2-T)^3 = (3-T)(T^2-3T+3),$$

which gives (9). $\square$

Proof of Theorem 1.1. Differentiating $T = 1 + tT^3$ and simplifying (9) gives

$$P(t) + Q(t) = 2T^9 + \frac{1}{3}t \frac{d}{dt} T^9.$$

Hence

$$[t^m](P(t) + Q(t)) = \frac{m+6}{3}[t^m]T^9.$$

The ordinary Lagrange inversion formula yields

$$[t^m]T^9 = \frac{9}{2m+9} \binom{3m+8}{m}.$$

For $m = 0$ this is immediate; for $m > 0$ it follows from

$$[t^m](1+U)^9 = \frac{9}{m}[u^{m-1}](1+u)^{3m+8}.$$

Consequently

$$[t^m](P(t) + Q(t)) = \frac{3(m+6)}{2m+9} \binom{3m+8}{m}.$$

The adjacent-binomial identity

$$\binom{3m+8}{m-1} = \frac{m}{2m+9} \binom{3m+8}{m}$$

gives (1). Substituting $m = n - 5$ proves Theorem 1.1. $\square$

The second main result evaluates the complete first-defect diagonal.

Theorem 6.2 (First defect diagonal). *Let $T = T(z) \in \mathbb{Q}[[z]]$ be the unique series with constant coefficient 1 such that $T = 1 + zT^3$, and put*

$$D = (3 - 2T)(T^2 - 3T + 3).$$

Then

(11)

$$\sum_{m \geq 0} T_{103}(m+4, 2m+4)z^m = \frac{T^3}{D}(9 - 6T + 27T^2 - 35T^3 + 11T^4 + 2T^5 - T^6).$$

7. FINITE DEFECTS AND THE FIRST DEFECT DIAGONAL

We use an integer-coordinate realization of the hexagonal grid. A cell is a pair $(i, j) \in \mathbb{Z}^2$, with $\kappa = 1 - i - j$. The (a, b) -benzel is

$$(12) \quad \mathcal{B}(a, b) = \{(i, j) : 1 - a \leq j - i, \kappa - j, i - \kappa \leq b - 1\}.$$

This is equivalent to the geometric definition in [3].

Put $n = a - 2$. An *owner anchor* is a pair (q, r) satisfying

$$q - r \equiv n \pmod{3}.$$

It owns the three labelled cells

$$(q, r)_0, \quad (q + 1, r)_1, \quad (q, r + 1)_2.$$

Every grid cell has exactly one owner and one label. For an owner meeting the benzel, define

$$(13) \quad u = \frac{n - q + r}{3}, \quad v = \frac{n - q - 2r}{3}, \quad w = \frac{n + 2q + r}{3}.$$

Then $u, v, w \in \mathbb{N}$ and $u + v + w = n$. Conversely,

$$q = w - u, \quad r = u - v,$$

so the owner is recovered from its simplex point.

The three cells owned by (u, v, w) have difference triples

label	$j - i$	$\kappa - j$	$i - \kappa$
0	$3u - n$	$3v - n + 1$	$3w - n - 1$
1	$3u - n - 1$	$3v - n$	$3w - n + 1$
2	$3u - n + 1$	$3v - n - 1$	$3w - n$

Substitution into (12) gives the following truncation.

Lemma 7.1 (Owner-domain truncation). *Suppose $d = 3k$ or $d = 3k + 1$, the benzel parameters are admissible, and $2k \leq n + 1$. The owners meeting $\mathcal{B}(a, 2a - d)$ are exactly*

$$\Delta_n^{(k)} = \{(u, v, w) \in \mathbb{N}^3 : u + v + w = n, \max(u, v, w) \leq n - k + 1\}.$$

Each of the three boundary segments $u = n - k + 1$, $v = n - k + 1$, and $w = n - k + 1$ contains k deficient owners. If $d = 3k$, each deficient owner contains two cells; if $d = 3k + 1$, it contains one cell.

Proof. For $d = 3k$, label 0 is present exactly when

$$u, w \leq n - k + 1, \quad v \leq n - k,$$

and the other two conditions are obtained cyclically. For $d = 3k + 1$, label 0 is present exactly when

$$u, v \leq n - k, \quad w \leq n - k + 1,$$

again with cyclic analogues. Taking the union over labels gives $\Delta_n^{(k)}$. On, for example, $u = n - k + 1$, the remaining coordinates are $(v, w) = (j, k - 1 - j)$ for $0 \leq j < k$, proving the boundary count and the claimed multiplicities. $\square$

Define the label potentials at owner (q, r) by

$$(15) \quad h_0(q, r) = q + r, \quad h_1(q, r) = -q, \quad h_2(q, r) = -r.$$

Their sum at a full owner is zero. A direct check on the four allowed prototiles gives the local energy table

(16)	tile class	in-phase stone	wrong-phase stone	two-owner bone	three-owner bone
	energy	0	3	1	4.

A right stone is *in phase* when it is one owner triple; otherwise it is wrong phase. A bone meets either two or three owners.

Theorem 7.2 (Finite-defect theorem). *Put $n = a - 2$, and suppose $k \geq 1$ and $2k \leq n + 1$. Assume that $\mathcal{B}(a, 2a - d)$ has a type-103 tiling. Let s be the number of wrong-phase stones and f the number of three-owner bones.*

- (1) *If $d = 3k$, then $s + f = \binom{k}{2}$.*
- (2) *If $d = 3k + 1$, then $s + f = \binom{k+1}{2}$.*

Proof. Along one deficient boundary, the sum of the potentials of its present cells is

$$\sum_{j=0}^{k-1} (n - 2k + 2 + j) = k \left(n - \frac{3(k-1)}{2} \right).$$

The three cyclic boundary sums are equal. Hence the energy of the entire region is

$$(17) \quad E_{a,d} = 3k \left(a - \frac{3k+1}{2} \right).$$

Using the benzel area and Conway-Lagarias invariant from [3], the number B of bones in any type-103 tiling is

$$(18) \quad B = \begin{cases} 3k(a - 2k), & d = 3k, \\ 3k(a - 2k - 1), & d = 3k + 1. \end{cases}$$

Here is the substitution explicitly. In terms of $n = a - 2$, the area formula gives the following numbers N of trihexes, while the invariant gives the

numbers S of right stones:

	N	S
$d = 3k$	$\frac{n^2 + 3n + 2 - 3k^2 + k}{2}$	$\frac{n^2 + 3n + 2 + 9k^2 - 6kn - 11k}{2}$
$d = 3k + 1$	$\frac{n^2 + 3n + 2 - 3k^2 - k}{2}$	$\frac{n^2 + 3n + 2 + 9k^2 - 6kn - 7k}{2}$

Since every tile has three cells and no left stones are allowed, $B = N - S$; this is exactly (18). Finally, summing (16) over the tiling gives

$$E_{a,d} = B + 3(s + f).$$

Substitution of (17) and (18) yields the two formulas. $\square$

Remark 7.3. If $d = 3k + 2$, all three labels at an owner have the same presence condition $\max(u, v, w) \leq n - k$. The resulting all-right-stone tiling is the unique stones-and-bones tiling by [3, Theorem 1.1]. This known residue class is not part of the new defect count in Theorem 7.2.

The offsets $d = 0, 1, 2$ have the unique all-right-stone tilings described in [3, Theorem 1.1 and Remark 5.1]; in particular they have no defect. For $d = 3$, Theorem 7.2 again gives zero defects. Thus $d = 4$ is the first diagonal with a nonzero defect count, and that count is exactly one. We now classify and enumerate this one-defect case.

8. THE ONE-DEFECT PATH BIJECTION

Set

$$a = m + 4, \quad b = 2m + 4, \quad n = m + 2.$$

The owner domain is the full simplex Δ_n , except that each of its three corner owners contains a single boundary cell. Every tiling has one bad tile, of one of five kinds: two wrong-stone phases or one of the three bone orientations in its three-owner phase.

The five possibilities admit the following uniform parameterization. Write $p = (x, y, z)$ and let c_ℓ be the owner of the label- ℓ cell of the bad tile. The condition $p \in \Delta_r$ means $x + y + z = r$.

(19)

class	parameter	c_0	c_1	c_2
S_1	$p \in \Delta_m$	$(x + 1, y, z + 1)$	$(x + 1, y + 1, z)$	$(x, y + 1, z + 1)$
S_2	$p \in \Delta_{m+1}$	$(x, y, z + 1)$	$(x + 1, y, z)$	$(x, y + 1, z)$
B_A	$p \in \Delta_m$	$(x, y, z + 2)$	$(x + 1, y + 1, z)$	$(x, y + 1, z + 1)$
B_B	$p \in \Delta_m$	$(x + 1, y, z + 1)$	$(x + 1, y + 1, z)$	$(x, y + 2, z)$
B_C	$p \in \Delta_m$	$(x + 1, y, z + 1)$	$(x + 2, y, z)$	$(x, y + 1, z + 1)$

Here S_1, S_2 are the two wrong-stone phases, and B_A, B_B, B_C are the three three-owner bone classes. These formulas follow by applying (13) to the three cells of each translated prototile; in particular, they are literal placement coordinates rather than a quotient or orbit parameterization.

Delete the bad tile. Every remaining bone is a two-owner bone; orient it from the owner containing two of its cells toward the owner containing its third cell. Label the edge by that third cell's label. We call the resulting directed graph the *owner graph*.

In owner-anchor coordinates, put

$$\alpha = (2, -1), \quad \beta = (-1, -1), \quad \gamma = (-1, 2).$$

A label-0 edge has step α or γ , a label-1 edge has step β or γ , and a label-2 edge has step α or β . Equivalently, in simplex coordinates these steps are

$$\alpha = (-1, 0, 1), \quad \beta = (0, 1, -1), \quad \gamma = (1, -1, 0).$$

An *admissible label- ℓ path* is a directed path from c_ℓ to the one-cell boundary owner of label ℓ , using the corresponding two steps and remaining in Δ_n . The path with no edges is allowed exactly when its two endpoints coincide.

Lemma 8.1 (Three paths). *The owner graph is the disjoint union of three directed paths. For each $\ell \in \{0, 1, 2\}$, the label- ℓ path begins at the owner of the label- ℓ cell of the bad tile and ends at the one-cell boundary owner of label ℓ . A path may be empty when its initial owner is already its boundary endpoint.*

Proof. At a full owner away from the defect, the local exact-cover alternatives are: an in-phase stone and no incident edge; one incoming and one outgoing edge of the same label; or three incoming edges, one of each label. This is the complete six-class check supplied by the two allowed bone phases in each orientation. Thus every owner incident to an edge is a pass-through vertex unless it is a source or terminal. At a defect owner distinct from its labelled boundary owner, the bad tile covers the label- ℓ cell and one outgoing label- ℓ edge covers the other two cells. At a boundary owner not already occupied by the bad tile, one incoming edge covers its unique cell. If a defect owner is already its matching boundary owner, these two incidences cancel and give the allowed empty path.

The label potentials in (15) strictly increase along directed edges, so directed cycles are impossible. Remove the empty paths from the following count. For a connected component C , let S_C be its number of defect sources, B_C its number of one-cell boundary terminals, and K_C its number of internal three-incoming terminals. Every remaining vertex has one incoming and one outgoing edge. Summing outdegree minus indegree over C therefore gives

$$S_C = B_C + 3K_C.$$

Globally the numbers of nonempty defect sources and nonempty boundary terminals are equal: both are three minus the number of empty paths. Hence $\sum_C K_C = 0$, so no internal three-incoming terminal occurs. With no merger and no directed cycle, every nonempty component is a single directed path with one defect source and one boundary terminal; an additional component would have neither a source nor a terminal and would therefore contain a directed cycle. Local exact coverage at a pass-through owner forces its

incoming and outgoing edges to carry the same label. The source and terminal labels consequently agree, proving the stated three paths. $\square$

The path steps have a useful monotonicity in simplex coordinates:

	label	nonincreasing	nondecreasing
(20)	0	w	v
	1	v	w
	2	v	w

For every one of the five defect classes, the three initial owners satisfy strict cyclic separations

$$(21) \quad w(c_1) < w(c_0), \quad u(c_2) < u(c_1), \quad v(c_0) < v(c_2).$$

The inequalities are read directly from the five rows of (19). For example, every row has $w(c_1) < w(c_0)$. A label-0 path can only increase w , whereas a label-1 path can only decrease it, so these two paths cannot meet. The other two inequalities give the same argument for labels (1, 2) using u and labels (2, 0) using v . Hence independently chosen paths of successive labels cannot share an owner.

Conversely, fix a bad tile and three admissible labelled paths. Install the bad tile, realize each path edge by its corresponding bone, and put the in-phase right stone at every unused full owner. At a nonterminal core owner, the bad cell and first edge cover all present cells; for an empty path, the bad tile covers the unique cell of the coincident core/boundary owner. At a pass-through owner, the incoming edge covers the path label and the outgoing edge covers the other two; at a nonempty boundary endpoint, the incoming edge covers the unique cell. Thus the construction is a literal exact cover. Its local description also shows that it inverts deletion of the bad tile.

Proposition 8.2 (Literal bijection). *For every $m \geq 0$, type-103 tilings of $\mathcal{B}(m+4, 2m+4)$ are in bijection with a choice of one of the five defect classes and three independently admissible labelled paths from its core owners to the corresponding boundary owners.*

9. BALLOT ENUMERATION

For $p, q \in \mathbb{N}$, put

$$(22) \quad R(p, q) = \binom{p+2q}{q} - \binom{p+2q}{q-1},$$

where the second term is zero when $q = 0$. By the reflection principle, $R(p, q)$ counts walks starting at height p , with q up-steps and $p+q$ down-steps, that never go below zero. Equivalently,

$$(23) \quad R(p, q) = [x^q](1-x)(1+x)^{p+2q}.$$

For a simplex point (u, v, w) , the three labelled path counts are cyclically $R(u, v)$, $R(v, w)$, and $R(w, u)$. Substituting the five core parameterizations into these three factors gives the next theorem.

Indeed, read a path backwards from its boundary endpoint. For label 0, the two reverse steps become the down- and up-steps of a ballot walk that starts at height u , has v up-steps and $u + v$ down-steps, and ends at height zero. Thus its count is $R(u, v)$. Cyclic permutation gives $R(v, w)$ and $R(w, u)$ for labels 1 and 2. Applying this rule to (19) gives

class	product of its three arm counts
S_1	$R(x + 1, y)R(y + 1, z)R(z + 1, x)$
S_2	$R(x, y)R(y, z)R(z, x)$
B_A	$R(x, y)R(y + 1, z)R(z + 1, x)$
B_B	$R(x + 1, y)R(y + 1, z)R(z, x)$
B_C	$R(x + 1, y)R(y, z)R(z + 1, x)$.

The last two rows are cyclic reindexings of the B_A sum.

Theorem 9.1 (Exact cyclic ballot sum). *For $m = a - 4$, define*

$$(24) \quad A_m = \sum_{x+y+z=m+1} R(x, y)R(y, z)R(z, x),$$

$$(25) \quad C_m = \sum_{x+y+z=m} R(x + 1, y)R(y + 1, z)R(z + 1, x),$$

$$(26) \quad H_m = \sum_{x+y+z=m} R(x, y)R(y + 1, z)R(z + 1, x).$$

Then

$$(27) \quad T_{103}(a, 2a - 4) = A_m + C_m + 3H_m.$$

Proof. The second wrong-stone phase is parameterized by $x + y + z = m + 1$ and gives (24); the first is parameterized by $x + y + z = m$ and gives (25). One three-owner bone orientation gives (26). The other two orientations yield cyclic permutations of its summand, hence the same total. Proposition 8.2 makes the three arm choices independent and proves that these five classes exhaust the literal tilings without multiplicity. $\square$

10. LAGRANGE-GOOD EVALUATION

We evaluate all coefficients at once. Let x_0, x_1, x_2 be commuting variables and read subscripts cyclically. Define

$$(28) \quad \phi_0 = (1 + x_0)^2(1 + x_1), \quad \phi_1 = (1 + x_1)^2(1 + x_2), \quad \phi_2 = (1 + x_2)^2(1 + x_0),$$

and $w_i = x_i/\phi_i$. The specialized Good determinant is

$$(29) \quad \Delta(x) = \frac{1 - x_0 - x_1 - x_2 + x_0x_1 + x_0x_2 + x_1x_2 - 2x_0x_1x_2}{(1 + x_0)(1 + x_1)(1 + x_2)}.$$

The multivariate Lagrange–Good formula [4, 5] applies to every formal series

$$F \in \mathbb{Q}[[x_0, x_1, x_2]]$$

and gives

$$(30) \quad F(x) = \sum_{\mathbf{r} \in \mathbb{N}^3} [x^{\mathbf{r}}] F(x) \phi(x)^{\mathbf{r}} \Delta(x) w(x)^{\mathbf{r}}.$$

For completeness, the determinant in (29) is obtained by expanding

$$\det \left(\delta_{ij} - \frac{x_j}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right)_{0 \leq i, j \leq 2}.$$

Introduce the three numerators

$$N_A = (1 - x_0)(1 - x_1)(1 - x_2),$$

$$N_C = (1 - x_0^2)(1 - x_1^2)(1 - x_2^2),$$

$$N_H = (1 - x_0^2)(1 - x_1)(1 - x_2^2),$$

and put

$$F_A = \frac{N_A}{\Delta}, \quad F_C = \frac{N_C}{\Delta}, \quad F_H = \frac{N_H}{\Delta}.$$

For $F = \sum_{i, j, k \geq 0} f_{i, j, k} x_0^i x_1^j x_2^k$, define its total-degree diagonal by

$$\text{Diag } F(z) = \sum_{r \geq 0} \left(\sum_{i+j+k=r} f_{i, j, k} \right) z^r.$$

Using (23), the coefficient in (30) is, respectively, the summand in (24), (25), or (26). Thus total-degree diagonalization of (30) gives the three component series.

To make this explicit, write

$$A_0(r) = \sum_{x+y+z=r} R(x, y) R(y, z) R(z, x)$$

and let $A_0(z) = \sum_{r \geq 0} A_0(r) z^r$, with analogous notation for $C(z)$ and $H(z)$. On the diagonal $x_0 = x_1 = x_2 = s$, all three w_i equal

$$(31) \quad q(s) = \frac{s}{(1+s)^3}.$$

The determinant becomes

$$(32) \quad \Delta(s, s, s) = \frac{(1-2s)(1-s+s^2)}{(1+s)^3}.$$

Consequently,

$$(33) \quad A_0(q(s)) = \frac{(1-s)^3(1+s)^3}{(1-2s)(1-s+s^2)},$$

$$(34) \quad C(q(s)) = \frac{(1-s)^3(1+s)^6}{(1-2s)(1-s+s^2)},$$

$$(35) \quad H(q(s)) = \frac{(1-s)^3(1+s)^5}{(1-2s)(1-s+s^2)}.$$

Now let $T = 1 + zT^3$ and set $s = T - 1$. Equation (31) gives $q(s) = z$. If

$$D = (3 - 2T)(T^2 - 3T + 3),$$

then (33)–(35) become

$$(36) \quad A_0(z) = \frac{T^3(2-T)^3}{D}, \quad C(z) = \frac{T^6(2-T)^3}{D}, \quad H(z) = \frac{T^5(2-T)^3}{D}.$$

Since $A_m = A_0(m+1)$ and $A_0(0) = 1$,

$$\sum_{m \geq 0} A_m z^m = \frac{A_0(z) - 1}{z}.$$

Combining this identity with (27) and (36), then using $z = (T-1)/T^3$, gives exactly the rational function in Theorem 6.2.

11. COMPUTATIONAL CHECKS

A direct exact-cover program implements the barycentric benzel and the four permitted prototile classes directly. It reproduces the formula for $5 \leq n \leq 11$ and records the owner graph of every tiling. A separate directed-path program verifies the fixed-sink chirality products and the total formula through $n = 12$. These computations are finite consistency checks and are not used in the proof of Theorem 1.1.

An accompanying Lean 4 development formalizes both enumerative arguments and the general finite-defect theorem directly over the literal benzel and exact-cover definitions. It includes the tiling/path equivalences, the one-defect classification, the ballot sums, and both generating-function evaluations. The checked public endpoints use no project-specific axioms or computational proof shortcuts; their reported logical dependencies are only the standard Lean/Mathlib principles `propext`, `Classical.choice`, and `Quot.sound`. The finite computations remain consistency checks rather than premises of the proofs.

Preprint history. Earlier versions under concept DOI [10.5281/zenodo.22057378](https://doi.org/10.5281/zenodo.22057378) contained only the solution of Propp’s Problem 6. The present article subsumes that version and adds the finite-defect hierarchy and the complete first-defect diagonal.

Data and code availability. The released Problem 6 formalization and reproducibility files are available in the [public code repository](#); the versioned software archive has DOI [10.5281/zenodo.22079396](#). The combined source and $d = 4$ formalization are included in the versioned releases of the same repository.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author used an author-developed local implementation of the Proof Engine framework for AI-assisted mathematical research. The implementation operates through replaceable AI-agent and harness components; the present work used OpenAI Codex within that implementation. The framework is described in *Proof Engine Infrastructure* [6]. The author reviewed and edited the resulting material and assumes full responsibility for the definitions, statements, proofs, citations, code, and exposition.

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